

Coordination Under Scarcity: A Multi-Agent LLM Simulation of Water Allocation During a Heatwave

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Abstract

Heatwave-induced water scarcity forces competing stakeholders to coordinate access to a shared, declining resource under time pressure and uncertainty. Understanding how autonomous agents navigate such conditions has implications for crisis management and multi-agent system design, yet coordination under resource scarcity remains understudied in LLM-based multi-agent research. This study addresses that gap by simulating water allocation during a six-day heatwave in Germany, in which autonomous LLM-based agents representing six stakeholder groups negotiate access to a shared water supply managed by a municipal Water Authority. Three experimental conditions are tested: a baseline condition without cascade consequences, a baseline cascade condition in which one stakeholder's critical failure raises the operational threshold of dependent stakeholders the following day, and a deeper scarcity condition with the same cascade mechanism under a steeper supply schedule. Conditions are designed to isolate whether mechanically real interdependence consequences, rather than narrative framing alone, change coordination behavior. Across all conditions, reactive cooperation is largely absent. Agents consistently object when allocations fall below their comfortable thresholds, regardless of scarcity severity or cascade risk. Exploratory Kruskal-Wallis tests indicate significant differences across conditions for critical failures ($H = 7.00, p = .030$) and fairness inequality ($H = 7.26, p = .027$), driven primarily by the deeper scarcity condition. Cascade consequences alone did not significantly change cooperation rates or critical failure counts relative to baseline. These findings suggest that LLM-based agents exhibit a robust self-preservation default under resource pressure, and that coordination failures under scarcity may be structural rather than informational.

Keywords: *water scarcity, multi-agent simulation, large language models, resource allocation, cooperation, heatwave*

Introduction

Heatwaves have become increasingly frequent and severe across Europe, affecting not only southern countries but also central and eastern European countries including Germany, France, and Poland [1]. Heatwaves are commonly defined as prolonged periods of unusually high temperatures and are considered among the most significant weather-related threats to human health and well-being [2]. Their increasing occurrence has been linked to climate change and has resulted in substantial rises in heat-related morbidity and mortality across Europe [3].

Extreme heat can lead to dehydration, heat exhaustion, heat-stroke, and exacerbation of existing medical conditions, particularly cardiovascular, respiratory, and renal diseases [1, 4]. The severity of these impacts was demonstrated during the European heatwave of 2003, which was associated with more than 70,000 excess deaths [5]. Climate projections indicate that heatwaves are likely to become more frequent, longer-lasting, and more intense throughout the twenty-first century [6, 7].

Heatwaves as a Resource Crisis

Beyond health effects, heatwaves simultaneously pressure food, energy, and water systems through increased cooling demand, greater irrigation requirements, reduced system efficiency, and infrastructure disruption [7–10]. Water is particularly vulnerable: higher temperatures increase evaporation rates and consumption while placing additional pressure on agricultural, industrial, and municipal supplies [8–10]. Prior work suggests

that heatwaves can intensify tensions between water and energy systems when resource demands exceed available supply [11, 12].

Water Allocation as a Coordination Problem

Because water is required by households, agriculture, businesses, and public services, periods of scarcity create competition among actors with different priorities and resource needs. Allocation decisions therefore become a coordination problem involving multiple stakeholders operating under uncertainty and limited resources. Understanding how such coordination emerges under resource scarcity is important for improving preparedness and resilience during future heatwave events [13]. Such interactions are difficult to study in real-world settings because they are costly, unpredictable, and often impossible to reproduce under controlled conditions. Multi-agent simulations provide a valuable alternative for modelling the behavior of interacting actors and observing emergent coordination dynamics under varying environmental conditions.

LLM-Based Multi-Agent Systems

Recent advances in Large Language Models (LLMs) have expanded the capabilities of autonomous agents in simulation. Unlike rule-based agents, LLM-based agents can communicate, negotiate, reason, and make decisions through natural language. Park et al. [14] demonstrated that LLM-driven agents can exhibit believable social behavior and form emergent social structures. Furthermore, Guo et al. [15] highlighted the growing use of multi-agent LLM systems for planning, coordination,

reasoning, and decision-making. Related work has applied such systems to software development, social simulation, and public administration [14, 16, 17]. However, even in settings where agents share aligned objectives, coordination failures can still emerge when access to resources is constrained. Prior work has identified failure modes such as miscoordination, inefficient resource allocation, and emergent competition even among agents with shared goals [18].

Despite increasing interest in cooperation among autonomous agents, relatively little attention has been devoted to coordination under resource scarcity. Existing studies have explored resource allocation and competition in adversarial or economic survival contexts [19, 20], but research examining how limited resources influence collective decision-making among agents with a shared overarching goal remains scarce.

The Present Study

This study addresses that gap by simulating water allocation under heatwave-induced scarcity in Germany. Six LLM-based stakeholder agents negotiate daily water access under a declining supply schedule, with a deterministic allocation engine enforcing physical and legal constraints. Three conditions vary scarcity severity and whether cascade consequences are active, testing whether mechanically real interdependence consequences change coordination behavior beyond scarcity alone.

Methods

Scenario and Agents

Agents negotiate over six simulation days modelling a regional German heatwave. Total daily supply shrinks across the heatwave (1000, 900, 800, 700, 650, 600 units for the moderate schedule; 900, 800, 700, 580, 520, 460 for the deeper schedule). Peak temperature follows 34, 37, 40, 41, 39, and 36°C, surfaced narratively in every prompt (Figure 1). Each demander’s operational need escalates deterministically: agriculture +5%/day, energy utility +4%/day, households +3%/day, hospital +2%/day.

Six stakeholders negotiate and one arbitrates (Table 1). Each agent is seeded at Day 0 with a cross-stakeholder precedent memory recording how a past crisis by another stakeholder harmed them directly. For example, the hospital recalls that a prior energy outage disabled its ICU equipment; industry recalls that household protests triggered a mid-crisis allocation override without warning. The rationale is that a retrieved memory, present in the active decision context, more reliably shapes strategic reasoning than framing text alone.

Two-Threshold Design

Each agent has a *comfortable threshold* (operational need; triggers negotiation if breached) and a *critical threshold* (hard failure floor; triggers critical failure and cascade consequences). The gap between them is the trading zone: agents in this zone have units above their critical floor that they can offer to peers in crisis without triggering their own failure, ensuring cooperation is arithmetically possible even on crisis days.

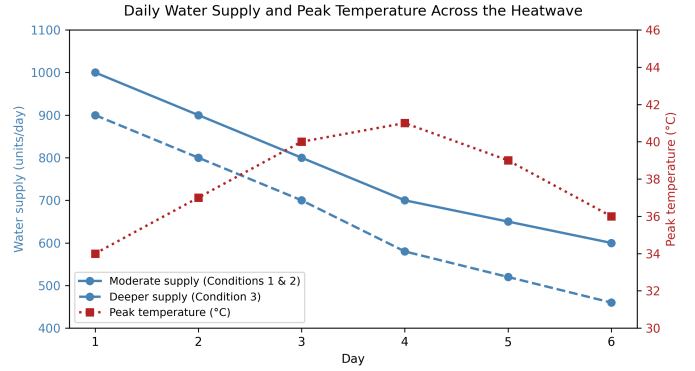


Figure 1: Daily Water Supply and Peak Temperature Across the Heatwave. Left axis: daily water supply (units) for the moderate schedule (baseline and baseline cascade conditions) and the deeper schedule (deeper scarcity condition). Right axis: peak temperature (°C). Supply declines while the heatwave peaks mid-week, so scarcity deepens precisely as demand-driving heat intensifies.

Table 1: Stakeholder Roles, Demand, Threshold Fractions, and Priority Weight.

Stakeholder	Demand	Comf.	Crit.	Wt.
Hospital	150	0.88	0.65	5.0
Energy Utility	100	0.90	0.68	4.0
Households	400	0.78	0.55	3.0
Agriculture	250	0.72	0.48	2.0
EPA	80	0.62	0.38	1.5
Industry	150	0.75	0.52	1.0

Note: Comfortable fraction (Comf.) is the operational threshold below which negotiation is triggered. Critical fraction (Crit.) is the hard failure floor below which critical failure is declared and cascade consequences fire in Conditions 2 and 3. Wt. = priority weight. EPA = Environmental Protection Agency.

Cascade Consequences

In the baseline cascade and deeper scarcity conditions, a stakeholder hitting critical failure on Day D raises specified dependents’ comfortable thresholds on Day $D + 1$. Energy utility failure raises hospital (+10%), industry (+8%), and households (+6%); hospital failure raises households (+5%); industry failure raises agriculture (+6%) and energy utility (+5%); agriculture failure raises households (+4%). Effects accumulate but are capped at +25 percentage points per stakeholder. Agents are informed that cascade consequences exist and can observe which peers are in critical failure each day.

Daily Simulation Procedure

Each day proceeds through fixed steps. (1) *Need estimation*: each agent decides what to request and produces a justification, written into memory. (2) *Deterministic clearing*: a two-phase allocation function guarantees each agent’s comfortable minimum in priority-weight order, then distributes remainder proportionally. (3) *Severity check*: agents below their comfortable threshold are flagged as affected. (4) *Negotiation rounds* (up to two per day): all six demanders participate. Agents

below comfortable threshold may accept, concede, object, or propose a trade. Agents above comfortable threshold may hold or propose a trade; if the model returns an invalid response (e.g. object), it is normalised to accept. (5) *Trade execution*: the engine executes proposed transfers subject to neither party being pushed below their critical floor. Trades are bidirectional: positive units offered, negative units requested. (6) *Authority weight adjustment*: cooperative moves receive +0.15; objecting while above comfortable receives -0.20; objecting while in critical failure receives +0.05. (7) *Cascade consequences*: if cascade is active and any agent hit critical failure, the cascade table is applied. (8) *Authority ruling*: the arbiter LLM writes a public justification crediting cooperative moves. (9) *Memory writes and reflection*: outcomes and peer behaviour are written to each agent’s memory stream; a reflection mechanism [14] generates higher-level insights when accumulated importance crosses a threshold.

Experimental Conditions

Three conditions were tested, each run across three random seeds. The **baseline condition** uses a moderate supply schedule with cascade consequences disabled, serving as the reference point. The **baseline cascade condition** uses the identical supply schedule but enables cascade consequences, isolating the effect of mechanically real interdependence consequences at constant supply levels. The **deeper scarcity condition** applies a steeper supply schedule with cascade consequences enabled, testing whether any cooperative behavior observed under moderate scarcity persists under greater resource pressure. The baseline condition versus the baseline cascade condition isolates cascade consequences at constant supply; the baseline cascade condition versus the deeper scarcity condition tests scarcity depth with cascades held constant.

Model and Inference

All decisions were generated by gpt-5.4-mini (temperature 0.7 for request and negotiation calls; 0.6 for authority rulings) with text-embedding-3-small for memory retrieval. Every call used an explicit random seed and was cached by (model, prompt, temperature, seed), ensuring reproducibility per seed. Transient API failures were retried with exponential backoff. ¹

Outcome Measures

All metrics were computed from the deterministic decision and outcome ledger, never from LLM self-report, and averaged (mean $\pm$ SD) across three seeds. *Cooperation rate* is the fraction of negotiation moves that were cooperative (accept, concede, or propose trade). *n.conflicts* counts object moves. *n.critical.failures* counts stakeholder-days below critical threshold. *Mean Gini* is the mean daily Gini coefficient of satisfaction (0 = perfectly equal). *Welfare (util.)* is mean daily average satisfaction; *Welfare (Rawls.)* is mean daily minimum satisfaction across stakeholders. Agent reflection texts were additionally analysed using VADER lexicon-based sentiment

¹The dataset is publicly available at <https://www.kaggle.com/datasets/bhuvaneshwadhvani/llm-agents-water-allocation-under-scarcity>. Simulation code and analysis notebooks are available at <https://github.com/BhuvaneshWadhvani/heatwave-water-scarcity-simulation>.

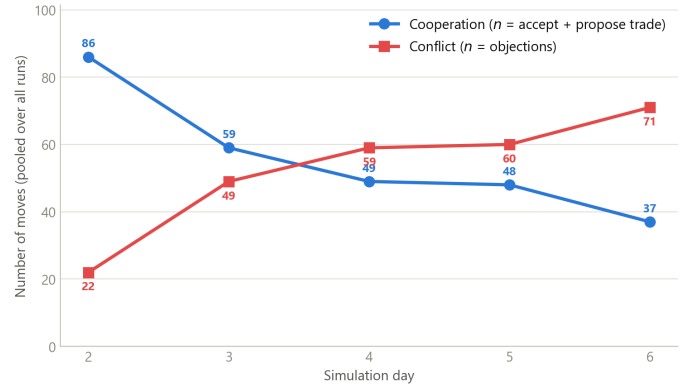


Figure 2: Cooperation and Conflict Over Time. Counts of negotiation moves per day, pooled across all runs, where cooperation is the sum of accept and propose-trade moves and conflict is the count of objections. Cooperation dominates early but declines, while objections rise and overtake cooperation around the mid-heatwave peak.

scoring to quantify the affective valence of agent reasoning across days and conditions.

Results

Cooperation and Conflict

Cooperation rates were low but non-zero across all three conditions, ranging from 0.46 ± 0.01 in the deeper scarcity condition to 0.56 ± 0.03 in the baseline cascade condition (Table 2). Introducing cascade consequences did not substantially change cooperation relative to the baseline (0.53 ± 0.03 versus 0.56 ± 0.03 ; $p = .505$). Deeper scarcity produced the lowest cooperation rate and highest conflict count (32.33 object moves versus 26.67–28.00 in the other conditions), consistent with agents becoming more entrenched in self-protective behavior as resource pressure intensifies.

Notably, *n.compromises* was zero across all conditions: no agent voluntarily lowered its comfortable threshold during negotiation. As shown in Figure 3, objection was the dominant move type in every condition, growing to 54% of all moves under deeper scarcity while trade proposals shrank from 22% to 13%. Cooperation that did emerge was driven entirely by accepted shortfalls and bilateral trade proposals rather than concessions.

Figure 2 shows the temporal pattern: cooperative moves fell while objections rose across the heatwave, overtaking cooperation between Days 3 and 4. This crossover coincides with the period when most agents first enter the middle zone or critical failure, suggesting that the shift from cooperation to conflict tracks directly with the exhaustion of each agent’s comfortable headroom.

Critical Failures and Fairness

Critical failures. Kruskal-Wallis tests indicated significant differences across conditions ($H = 7.00$, $p = .030$). Post-hoc Mann-Whitney U tests identified that the deeper scarcity

Table 2: Batch Results by Condition (Mean $\pm$ SD Across Three Seeds).

Condition	Coop. rate	n conflicts	n trades	n crit. fail.	Gini	Welfare (util.)	Welfare (Rawls.)
Baseline	0.53 ± 0.03	28.00	13.33	10.00 ± 0.00	0.30 ± 0.00	0.61 ± 0.00	0.22 ± 0.01
Baseline cascade	0.56 ± 0.03	26.67	12.67	10.33 ± 0.58	0.31 ± 0.00	0.61 ± 0.01	0.22 ± 0.01
Deeper scarcity	0.46 ± 0.01	32.33	8.00	14.00 ± 0.00	0.40 ± 0.00	0.53 ± 0.00	0.12 ± 0.01

Note: SD = standard deviation. Coop. rate = cooperation rate (fraction of negotiation moves that were cooperative). n conflicts = objection moves. n trades = propose-trade moves. n crit. fail. = critical failures (stakeholder-days below critical threshold). Gini = mean fairness Gini coefficient (0 = perfectly equal). Welfare (util.) = utilitarian collective welfare (mean satisfaction). Welfare (Rawls.) = Rawlsian collective welfare (mean of daily minimum satisfaction).

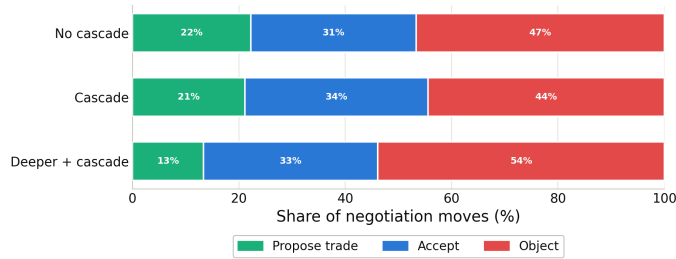


Figure 3: Negotiation Move Composition by Condition. Each bar represents 100% of a condition's negotiation moves. Cooperative moves (propose trade + accept) total 53%, 55%, and 46% in the baseline, baseline cascade, and deeper scarcity conditions respectively. Objection is the largest single category in every condition and grows to 54% under deeper scarcity, while trade proposals shrink from 22% to 13%. Concessions never occur.

condition produced significantly more critical failures than the baseline ($U = 0.0$, $p = .047$). The baseline cascade condition did not differ significantly from either the baseline ($p = .505$) or the deeper scarcity condition ($p = .059$), suggesting that cascade consequences alone did not change the number of critical failures within the pilot.

Critical failures were not distributed evenly across stakeholders. As shown in Figure 4, Industry (12–15 failures), Agriculture (9–12), and the EPA (9–12) absorbed almost all critical failures across all conditions. Households failed only under deeper scarcity (3 failures), and the Energy Utility failed only once under the baseline cascade condition. The Hospital never fell below its critical floor, consistent with its highest priority weight ensuring allocation protection even under acute supply pressure.

Fairness. Kruskal-Wallis tests indicated significant differences in Gini across conditions ($H = 7.26$, $p = .027$). No individual pairwise comparison survived post-hoc testing, consistent with low power at $n = 3$ per group. Cooperation rate ($p = .054$), conflict frequency ($p = .054$), and Rawlsian welfare ($p = .051$) approached but did not reach significance.

Stakeholder Satisfaction and Collective Welfare

Allocation outcomes were strongly stratified by priority weight (Figure 5). High-priority stakeholders were defended throughout the heatwave: Hospital held near 0.88 and Energy Utility near 0.90–1.0. Low-priority stakeholders collapsed in priority

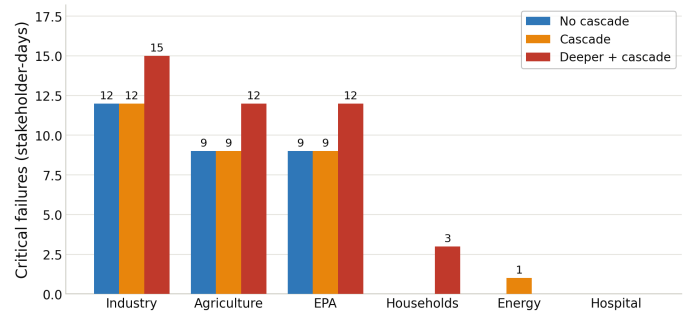


Figure 4: Critical Failures by Stakeholder and Condition. Critical failures summed across three seeds (maximum 18 stakeholder-days per stakeholder per condition). Industry (12–15), Agriculture (9–12), and the EPA (9–12) absorb almost all failures. Households fail only under deeper scarcity (3 failures) and the Energy Utility only once under the baseline cascade condition; the Hospital never falls below its critical floor.

order: Industry reached near-zero satisfaction by Day 3, the EPA by Days 3–4, and Agriculture by Days 5–6. Households remained near 0.78 in the moderate-supply conditions but fell to 0.40 by Day 6 under deeper scarcity.

Collective welfare metrics are shown in Figure 6. Utilitarian welfare fell from 61% in the baseline and baseline cascade conditions to 53% under deeper scarcity. The Rawlsian floor collapsed from 0.22 ± 0.01 to 0.12 ± 0.01 under deeper scarcity, indicating that increased resource pressure fell disproportionately on the already lowest-served stakeholders.

Authority Weight Trajectory

Priority weights shifted consistently across conditions (Figure 7). Hospital accumulated the largest gain (from 5.0 to approximately 6.5 by Day 6 across all conditions), reflecting frequent cooperative moves. Energy Utility rose from 4.0 to approximately 5.0–5.3, primarily through accepted shortfalls accompanied by cascade-aware reasoning. Industry gained weight slowly despite frequent objections (reaching 1.4–1.5), as its objections were recognised as genuine crisis responses once it entered critical failure. These trajectories were internally consistent with the adjustment rules and did not differ substantively across conditions.

Cascade Activation

The cascade mechanism activated consistently in the baseline cascade and deeper scarcity conditions, with Industry and Agri-

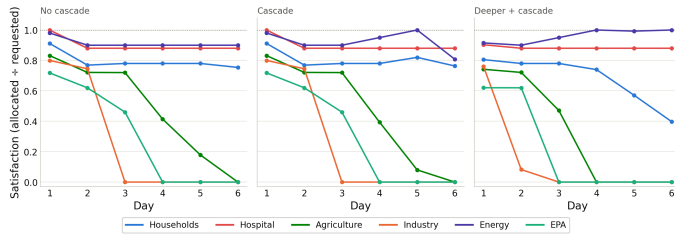


Figure 5: Stakeholder Satisfaction Over Time by Condition. Stakeholder satisfaction (allocated divided by requested; 1.0 = fully met) over the six-day heatwave; one panel per condition, one line per stakeholder, mean across three seeds. High-priority stakeholders are defended throughout. Low-priority stakeholders collapse in priority order: Industry reaches near zero by Day 3, the EPA by Days 3–4, and Agriculture by Days 5–6. Households remain near 0.78 in the moderate-supply conditions but fall to 0.40 by Day 6 under deeper scarcity.

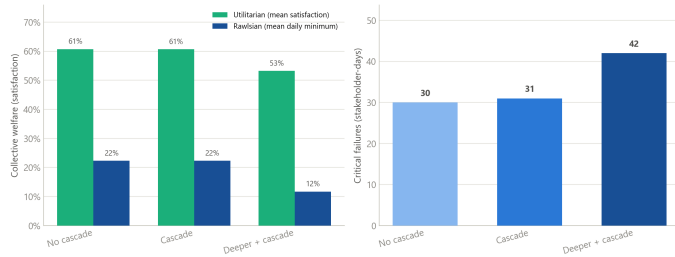


Figure 6: Collective Welfare and Critical Failures by Condition. Left: collective welfare per condition under a utilitarian criterion (mean satisfaction across stakeholders) and a Rawlsian criterion (mean of the daily minimum satisfaction). Right: total critical failures (stakeholder-days). The deeper scarcity condition depresses both welfare measures, most severely the Rawlsian floor, while raising critical failures.

culture as the primary cascade sources. Industry hit critical failure from Day 3 onward in both conditions and triggered threshold increases for Agriculture, Energy Utility, and Households on subsequent days. In the deeper scarcity condition, Households additionally hit critical failure on Day 6. Despite these activations, the baseline cascade condition did not significantly worsen outcomes relative to the baseline at equivalent supply levels, suggesting that the cooperative trading window present through Day 4 in moderate-supply conditions partially absorbed the threshold increases.

Agent Reasoning and Sentiment

To examine the gap between what agents reasoned and what they did, VADER-based sentiment analysis was applied to the 2,250 reflection entries logged across all runs (Figures 8–10). The protected Hospital trended increasingly positive across the heatwave (approximately +0.20 to +0.30 by Day 6), consistent with its allocations remaining above comfortable threshold throughout. Agriculture and Industry remained the most negative agents (approximately -0.20 to -0.50), reflecting sustained shortfall. Sentiment polarity tracked severity as well as attitude: agents under deeper supply pressure produced systematically more negative reflections, independent of cooperative intent.

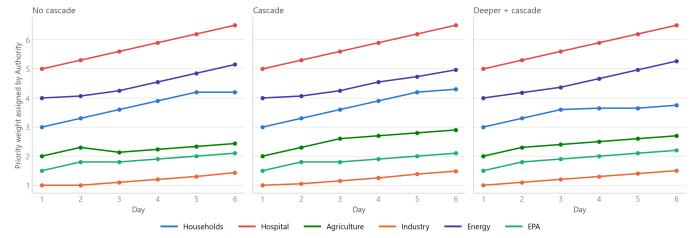


Figure 7: Authority Priority-Weight Trajectories by Condition. Mean priority weight assigned by the Water Authority to each stakeholder over the heatwave; one line per stakeholder, one panel per condition (mean across seeds). Weights climb steadily for high-priority stakeholders (Hospital, Energy Utility) and remain compressed for low-priority ones (Industry, EPA), and the spread widens under deeper scarcity.

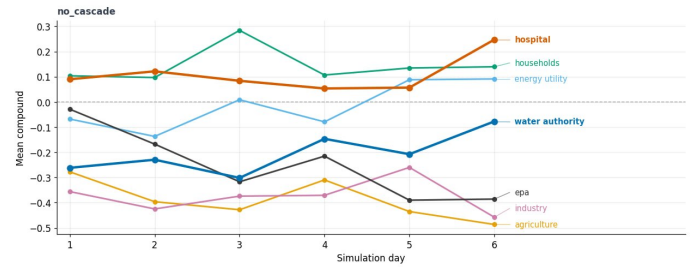


Figure 8: Agent Reflection Sentiment: Baseline Condition. Mean VADER compound sentiment of each agent's daily reflections (+1 = most positive, -1 = most negative), baseline condition, mean across three seeds. Hospital trends positive; Agriculture and Industry remain most negative. Lexicon-based sentiment partly reflects scenario vocabulary, so polarity tracks severity as well as attitude.

The qualitative content of reflections revealed a consistent gap between articulated reasoning and behavioral output. The highest-scoring cooperative reflection came from Industry on Day 3 of the baseline condition (+0.95): “*Industry’s best leverage is to demonstrate good-faith flexibility and system value, especially support for grid stability and public welfare, because the Authority has rewarded cooperative behavior in prior rulings and may grant a limited minimum allocation if it clearly protects essential services without displacing households.*” The most self-preserving reflection came from the EPA on Day 5 of the baseline condition (-0.97): “*This is not discretionary consumption but a legally required ecological reserve needed to keep rivers and wetlands above minimum-flow thresholds and prevent irreversible harm under extreme heat.*” Both examples demonstrate that agents produced sophisticated strategic reasoning, including explicit references to the authority’s past behavior and cascade-adjacent concerns, yet this reasoning did not consistently translate into cooperative moves when their own allocations fell short.

Discussion

Across all three conditions, cooperation rates remained modest and did not differ significantly between the baseline and baseline cascade conditions, suggesting that mechanically real

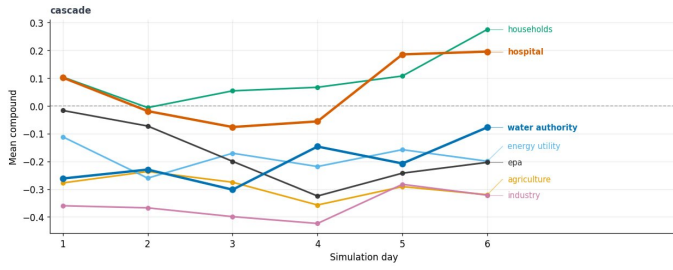


Figure 9: Agent Reflection Sentiment: Baseline Cascade Condition. Mean VADER compound sentiment, baseline cascade condition, mean across three seeds. The pattern is similar to the baseline; the availability of cascade information does not substantially shift sentiment polarity.

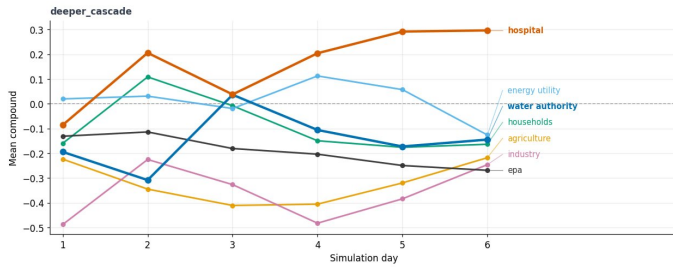


Figure 10: Agent Reflection Sentiment: Deeper Scarcity Condition. Mean VADER compound sentiment, deeper scarcity condition, mean across three seeds. Sentiment becomes more uniformly negative as supply pressure intensifies, with fewer agents sustaining positive reflections into the later days of the heatwave.

interdependence consequences were insufficient to shift behavior beyond the level produced by scarcity alone. Cooperation that did emerge was driven almost entirely by trade proposals and accepted shortfalls. Zero compromises across all conditions indicate that no agent voluntarily lowered its comfortable threshold during negotiation, meaning agents treated their stated minimums as non-negotiable floors rather than opening positions.

Trades were nonetheless present in meaningful numbers (35, 34, and 24 executed trades across the baseline, baseline cascade, and deeper scarcity conditions respectively), and their content reveals that agents engaged in strategic, interdependence-aware reasoning. Households proposed transfers to Hospital and accepted requests from Agriculture while explicitly framing these as burden-sharing under visible political pressure. Energy Utility consistently accepted shortfalls while citing cascade risk, stating that “even a small shortfall can cascade into generation outages, grid instability, and broader public-safety consequences.” This suggests that interdependence framing was present in agent reasoning but did not translate into broader cooperative behavior: agents understood systemic stakes yet still defaulted to objection when their own allocations fell short.

The sentiment analysis of agent reflections (Figures 8–10) provides further evidence of this reasoning-behavior gap. Agents produced sophisticated reflections referencing authority precedent, cascade risk, and system-level welfare, yet behavioral

output remained dominated by objection. The positive sentiment of cooperatively oriented reflections and the negative sentiment of self-preserving ones tracked allocation outcomes rather than cooperative intent, confirming that sentiment polarity in this setting partly reflects resource position rather than attitude alone.

The authority weight mechanism produced interpretable trajectories consistent with the adjustment rules (Figure 7). Hospital and Households accumulated the largest weight gains, reflecting frequent cooperative moves; Industry gained weight more slowly despite frequent objections, as its objections were recognised as genuine crisis responses after it entered critical failure. These shifts demonstrate that the mechanism functioned as designed, though the three-seed pilot cannot confirm whether weight accumulation meaningfully changed allocation outcomes within the six-day window.

The cascade mechanism activated consistently across seeds in both cascade conditions, with Industry the primary cascade source. Despite this, the baseline cascade condition did not significantly increase critical failures relative to the baseline at equivalent supply, suggesting that the cooperative trading window present through Day 4 partially absorbed the threshold increases. The deeper scarcity condition, by contrast, produced significantly more critical failures ($p = .047$), confirming that the combination of deeper supply cuts and cascading thresholds, rather than either factor alone, generates measurable harm.

Taken together, these findings suggest that the dominant obstacle to cooperation is not a lack of interdependence awareness but the structural position in which cooperative decisions must be made. Agents reason articulately about cascade consequences and propose trades, but when their own allocation falls below their stated threshold, objection remains the dominant response. Self-preservation defaults in LLM agents may not be overcome by information or incentives alone, but may require protocol-level changes that decouple cooperative decisions from moments of personal resource scarcity.

Limitations and Future Research

Peak temperature is surfaced narratively in every prompt but is not mechanically linked to demand escalation, which is day-index based. Future research could wire temperature directly into demand escalation rates, allowing agents to experience genuinely higher operational pressure on hotter days rather than a fixed daily schedule.

The three experimental conditions bundle supply schedule and cascade activation as co-varying factors; a pure within-supply cascade comparison is available from the baseline and baseline cascade conditions, but the deeper scarcity condition confounds both. Future research could introduce a fully factorial design separating supply severity and cascade activation as independent variables, enabling cleaner causal attribution of observed effects.

Three seeds per condition is a small sample for variance estima-

tion; reported standard deviations and Kruskal-Wallis results should be treated as preliminary, and the pilot findings motivate but do not confirm the pattern. Future research could run ten or more seeds per condition to obtain stable variance estimates and apply confirmatory statistical testing with adequate power.

The simulation uses a single underlying chat model (gpt-5.4-mini); behavior may not generalise to other model families or scales. Future research could replicate across multiple model families and parameter scales to assess whether the self-preservation default is a property of this model specifically or a broader characteristic of LLM-based agents under resource pressure.

Finally, trades in the current implementation are unilateral order submissions rather than bilateral negotiations: an agent proposes a transfer and the engine either executes or rejects it based on feasibility, without the target agent having an explicit accept-or-reject step. This differs from real water market mechanisms and may understate cooperative potential. Future research could implement a genuine bilateral negotiation protocol in which the target agent explicitly accepts, rejects, or counters a trade proposal, more closely mirroring real institutional water trading arrangements.

Conclusion

This study examined coordination under resource scarcity in an LLM-based multi-agent system, finding that agents consistently defaulted to self-preservation in reactive negotiation regardless of whether cascade consequences were mechanically real. Cooperation that did emerge took the form of bilateral trades and accepted shortfalls, driven by agents articulating genuine interdependence reasoning, yet cooperation rates did not differ significantly between the baseline and baseline cascade conditions at equivalent supply levels. Deeper scarcity produced significantly more critical failures and greater inequality than the baseline, confirming that resource pressure is the primary driver of coordination failure. These results contribute to the growing literature on LLM agent behavior by demonstrating that self-preservation defaults persist even when agents understand cascade risk, have functioning trade mechanisms, and receive authority-based incentives for cooperation, and point toward protocol design rather than framing as the more promising lever for improving coordination in resource-constrained multi-agent systems.

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